

- 2 (a) sum of random distribution of kinetic and potential energies of the molecules of the gas B1
- no intermolecular forces for ideal gas M1
- so no potential energy A1
- therefore, internal energy equals to total kinetic energy of the molecules only
- (b) (i) higher temperature increases r.m.s. speed and change in momentum of each molecule colliding with wall of container M1
- increase volume reduces frequency of collision M1
- force is product of frequency of collision and change in momentum which is constant when increase in change in momentum is balanced by reduction in frequency of collision A1
- (ii) increase in internal energy = $\frac{3}{2}k(460 - 280) \times 1.2 \times 6.02 \times 10^{23}$ (= 2692 J) C1
- heat supplied = $1.2 \times 0.032 \times c \times (460 - 280)$ (= 6.912c) C1
- heat supplied = increase in internal energy – work done on gas
 $6.912c = 2692 + 1300$ (correct use of first law) C1
- $c = 580 \text{ J kg}^{-1} \text{ K}^{-1}$ or $578 \text{ J kg}^{-1} \text{ K}^{-1}$ A1
- (c) constant volume so no work done B1
- less thermal energy needed for same temperature rise, so smaller specific heat capacity B1

3 (a) series of distinct coloured lines against dark background

B1